

续. 反函数定理证明.

下面来证明所得到的 x 满足 $f(x)=y$ 是唯一.

引理, 正则性引理. 若 $f: U \rightarrow V$ 是同胚, $f \in C^1(U, V)$, 则
 f 是微分同胚 $\Leftrightarrow x \in U, df(x): \mathbb{R}^n \rightarrow \mathbb{R}^n$ 可逆
 $\Updownarrow$
 $f' \in C^1$

$$\begin{aligned} \text{证(1). } \|df(x) - df(x_0)\| &\leq \varepsilon = \frac{1}{2\|df^{-1}(x_0)\|} \\ \Rightarrow \|df^{-1}(x_0)\| \cdot \|df(x) - df(x_0)\| &\leq \frac{1}{2} \\ \Rightarrow \underbrace{\|I - df^{-1}(x_0)df(x)\|}_{:=E} &\leq \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} \Rightarrow 1 + E + E^2 + \dots \text{ 收敛, 且收敛至 } (I - E)^{-1} &= (df^{-1}(x_0)df(x))^{-1} \quad \text{Neumann Series.} \\ \Rightarrow df(x) \text{ 可逆} \end{aligned}$$

$$\begin{aligned} \text{证(3). 取 } V &= B(y, \delta), \quad \delta = \frac{r}{4\|df^{-1}(x_0)\|} \\ U &= B(x_0, \frac{r}{2}) \cap f^{-1}(V). \end{aligned}$$

① 满:

$$\text{证(2). } y_1, y_2 \in V \quad x_1 = f^{-1}(y_1), \quad x_2 = f^{-1}(y_2).$$

$$\begin{aligned} \|x_1 - x_2\| &= \|T_{y_1}(x_1) - T_{y_2}(x_2)\| \\ &\leq \|T_{y_1}(x_1) - T_{y_1}(x_2)\| + \|T_{y_1}(x_2) - T_{y_2}(x_2)\| \\ &\leq \frac{1}{2}\|x_1 - x_2\| + \|df^{-1}(x_0)(y_1 - y_2)\| \\ &\leq \|df^{-1}(x_0)\| \|y_1 - y_2\| \end{aligned}$$

$$\Rightarrow \frac{1}{2}\|x_1 - x_2\| \leq \|df^{-1}(x_0)\| \|y_1 - y_2\|$$

$$\Rightarrow \|x_1 - x_2\| \leq K \|y_1 - y_2\| \Rightarrow \text{Lipschitz} \Rightarrow f^{-1} \in C(V)$$

□

C^m 反函数定理.

$\Omega \subset \mathbb{R}^n$ 开集, $f: \Omega \rightarrow \mathbb{R}^n$, $f \in C^m$ ($m \geq 2$)

$x_0 \in \Omega$, $df(x_0)$ 可逆, 则存在一个 x 的开邻域 U ,

以及 $f(x_0) = y$ 的开邻域 V , 使得 $f|_U: U \rightarrow V$ 是 C^m -微分同胚.

pf. 即证 $f^{-1} \in C^m$.

$g := (f|_U)^{-1} \in C^1$. $Jg(y) = (Jf(x))^{-1}$, 其中 $x = g(y)$.

令 $A(x) = Jf(x) = (A_{ij}(x))$ $A^{-1} = B$.

$B(x) = Jg(y) = (B_{kl}(y))$.

$$A_{ij}(x) = \frac{\partial f_i}{\partial x_j} \in C^{m-1}. \quad B_{kl}(y) = \frac{\partial g_k}{\partial y_l} = \frac{\det A^{(k,l)}}{\det A} \Big|_{x=g(y)} \in C^{m-1}$$

$\Rightarrow g$ 的 1 阶微分 $= C^{m-1}$ 函数 $\circ g$.

$$\textcircled{1} g \in C^1 \Rightarrow \text{RHS} \in C^1 \Rightarrow \frac{\partial g}{\partial y} \in C^1 \Rightarrow g \in C^2$$

$$\textcircled{2} g \in C^2 \Rightarrow g \in C^3$$

$\Rightarrow \dots$

$$\Rightarrow g \in C^m.$$

Ex. Kepler's Equation

$$M = E - e \sin E. \quad 0 \leq e < 1.$$

$$F(M, E) = E - e \sin E - M.$$

$$\frac{\partial F}{\partial E} = 1 - e \cos E > 0$$

$$\frac{dE}{dM} = \frac{\partial F / \partial M}{\partial F / \partial E} = \frac{1}{1 - e \cos E}.$$

定义. $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_m)$.

若系统 $F(x, y) = \begin{pmatrix} F_1(x, y) \\ \vdots \\ F_m(x, y) \end{pmatrix} = 0$ 决定 y 在 $U \subset \mathbb{R}^n$ 上作为 x 的函数, 即 $\forall x \in U, \exists! y \in \mathbb{R}^m$ s.t. $F(x, y) = 0$.

则称 $F(x, y)$ 定义了一个在 U 上隐函数 (implicit function) $y = f(x)$.

$$J_x F := \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \dots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}_{m \times n} \quad J_y F = \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \dots & \frac{\partial F_1}{\partial y_m} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial y_1} & \dots & \frac{\partial F_m}{\partial y_m} \end{pmatrix}_{m \times m}$$

$$JF = (J_x F, J_y F)_{m \times (n+m)}$$

定理. (隐函数定理)

Ω 是 $(x_0, y_0) \in \mathbb{R}^n \times \mathbb{R}^m$ 的一个邻域, $F: \Omega \rightarrow \mathbb{R}^m$ 满足:

1. $F \in C^k(\Omega, \mathbb{R}^m) \quad k \geq 1.$

2. $F(x_0, y_0) = 0$

3. $\det J_y F(x_0, y_0) \neq 0.$

则存在 (x_0, y_0) 的邻域 $I = I_{x_0} \times I_{y_0}$ 以及 $f \in C^k(I_{x_0}, I_{y_0})$, 使得

$F(x, f(x)) = 0.$ 以及 $Jf(x) = -(J_y F)^{-1} J_x F, \quad \forall x \in I_{x_0}.$

pf. 定义, $G: \Omega \rightarrow \mathbb{R}^n \times \mathbb{R}^m$

$$G(x, y) = (x, F(x, y))$$

$$JG = \begin{pmatrix} I_n & 0 \\ J_x F & J_y F \end{pmatrix} \quad \det JG = \det J_y F \neq 0$$

$G|_{\tilde{I}}:$

由反函数定理, $\overset{(x_0, y_0)}{\tilde{I}}$ 的邻域 $\tilde{I} \rightarrow V$ 是 C^k -微分同胚.

$$\Rightarrow \exists G^{-1}(x, y) = (x, \gamma(x, y)).$$

$$\text{取 } f = \gamma(x, 0).$$

Banach 空间 完备赋范线性空间.

$$\|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p \right)^{\frac{1}{p}}$$

$$\|x\|_{\infty} := \sup_{i \geq 1} |x_i|$$

Thm. 有限维赋范线性空间是 Banach 空间

L^p 空间.

$$T: X \rightarrow X$$

不动点定理 $\|T(x) - T(y)\| \leq q \|x - y\|, \quad 0 \leq q < 1.$

$$\Rightarrow T \text{ 有唯一不动点 } x^* \in X.$$